Disjoint Isolating Sets and Graphs with Maximum Isolation Number

Geoffrey Boyer and Wayne Goddard
School of Mathematical and Statistical Sciences, Clemson University

Abstract

An isolating set in a graph is a set X of vertices such that every edge of the graph is incident with a vertex of X or its neighborhood. The isolation number of a graph, or equivalently the vertex-edge domination number, is the minimum number of vertices in an isolating set. Caro and Hansberg, and independently Żyliński, showed that the isolation number is at most one-third the order for every connected graph of order at least 6. We show that in fact all such graphs have three disjoint isolating sets. Further, using a family introduced by Lemańska, Mora, and Souto-Salorio, we determine all graphs with equality in the original bound.

1 Introduction

Recently, Caro and Hansberg [3] introduced the concept of isolation in a graph G. Specifically, they considered a set X of vertices such that the graph $G - N[X]$ has some sparseness property, where $N[X]$ denotes the closed neighborhood of X (it together with its neighbors). For example, if $G - N[X]$ has no vertex, then X is a dominating set. The idea of "paying" for a vertex set but removing both it and its neighbors also appears in the vulnerability literature; see for example [4].

As a special case of the general concept, Caro and Hansberg [3] defined an *isolating set* of G as a set X of vertices such that $G - N[X]$ has no edge, and the *isolation number* of G, denoted $\iota(G)$, as the minimum cardinality of an isolating set. It should be noted that this parameter is equivalent to the *vertex-edge domination number* of G, as introduced earlier by Lewis et al. [9]. They defined a vertex u to *ve-dominate* an edge f if either f is incident to u, or at least one of the ends of f is adjacent to u. Further, a set S is *vertex-edge dominating* if every edge is *ve*-dominated by at least one vertex in S; in other words, all edges of G are incident to a vertex in $N[S]$, or equivalently, that S is an isolating set.

The isolation number is also the same as the 2-distance vertex covering number, as defined by Canales et al. [2].

Caro and Hansbeg proved a bound on the isolation number of a graph, later also proved by Żyliński for vertex-edge domination number:

Theorem 1 [3, 11] *If graph G is connected and not K_2 or C_5, then its isolation number is at most one-third its order.*

They and others posed the question of characterizing the graphs with isolation or vertex-edge domination number equal to one-third their order. Earlier Krishnakumari et al. [7] had characterized trees of order n with vertex-edge domination number equal to $n/3$, later also proved by Dapena et al. [5] for isolation number. Recently, Lemańska et al. [8] introduced an infinite family of graphs $\mathcal{G}$ such that every graph in the family has isolation number one-third its order. They showed that every unicyclic or block graph with this property is in this family, apart from two exceptions. They then suggested that apart from this family there is only a finite number of connected graphs in general with the property. This we prove in Section 3.

In another direction, it is well-known that every connected nontrivial graph has two disjoint dominating sets but not necessarily three. We show in Section 2 that every connected graph, except K_2 and C_5, has three disjoint sets such that the vertices not dominated by all three sets form an independent set. As a corollary this means that for a connected graph, except for K_2 and C_5, there are three disjoint (but not necessarily minimum) isolating sets. This strengthens a result of Kiser [6] (Theorem 4.8) who showed that every connected graph of order at least 3 has three disjoint distance-2 dominating sets. A *distance-2 dominating set* is a set X such that every vertex of the graph is either in $N[X]$ or has a neighbor in $N[X]$; this is a weaker property than isolating set.

2 Three Disjoint Sets that Nearly Dominate

We prove:

Theorem 2 *If G is a connected graph of order at least 3 other than C_5, then there exists a partition (A_1, A_2, A_3) of $V(G)$ such that $\mathcal{X} = \bigcup_{i=1}^{3} (V(G) - N[A_i])$ is an independent set.*

Note that the conclusion is slightly stronger than just $V(G) - N[A_i]$ being an independent set for each i.

We refer to the partition as a *coloring*. The proof of the theorem is by induction on the order and is divided into a sequence of claims. Furthermore, for the induction, note that the conclusion is also true for $G = K_1$ if one relaxes partition to weak partition. The base case is included in the following claim.

Claim 3 *If G is a star with at least three vertices, or a cycle C_n with $n \neq 5$, then there exists such a coloring.*

Proof. For the star, any coloring using all three colors is valid. So assume G is C_n and let r be the remainder when n is divided by 3. If $r = 0$ then use the coloring that repeats $1, 2, 3$; if $r = 1$ then use the coloring that is $1, 2, 1, 3$ followed by repeating $1, 2, 3$; and if $r = 2$ then use the coloring that is $1, 3, 2, 1, 3$ followed by repeating $1, 2, 3$. If $r = 0$ then $\mathcal{X}$ is empty; otherwise $\mathcal{X}$ contains two vertices and these are at distance at least two on the cycle. QED

So assume G is neither a star nor a cycle.

Claim 4 *If G has a cycle C of length a multiple of 3, then one can induct.*

Proof. Color the cycle C with the $1, 2, 3$ coloring as above. It follows that no vertex of C is in $\mathcal{X}$. Let F be a component of $G - C$. If F has a valid coloring then use it. If F is K_2, say x_1x_2 with x_1 having a neighbor v on C, color x_1 and x_2 so that x_1, x_2, and v have different colors. Only x_2 can be in $\mathcal{X}$. If F is C_5, say $x_1 \ldots x_5$ with x_1 having a neighbor v on C, then give x_3 the same color as v, give x_2 and x_5 the second color, and give x_1 and x_4 the third color. Only vertex x_5 can be in $\mathcal{X}$. QED

This idea can be re-used as follows:

Claim 5 *If G has a nontrivial path $P = v_1 \dots v_k$ such that $G-P$ is disconnected, and there exists a neighbor x of v_1 and a neighbor y of v_k in different components of $G-P$, then one can induct.*

PROOF. Color the path with the $1, 2, 3$ coloring. All vertices of P except v_1 and v_k are dominated by all three colors. Then color the remainder of the graph, as in the previous claim, choosing the color of x to be 3 and the color of y to be the color missing at v_k; hence no vertex of P is in $\mathcal{X}$. QED

Claim 6 *We may assume G is 2-connected and has minimum degree at least 3.*

PROOF. Suppose G has a cut-vertex v. Then since G is not a star, vertex v has a neighbor z in some nontrivial component of $G-v$. Then vz is a separating path, and one can apply the above claim, to induct. So we may assume G is 2-connected.

Suppose G has a vertex of degree 2. Consider the shortest cycle C containing such a vertex; necessarily induced. Since G is not a cycle, the cycle contains a vertex with a neighbor outside the cycle. Thus there must exist consecutive vertices v and w on C such that v has degree 2 in G and w has a neighbor outside the cycle. Let P be the path $C-v$; this satisfies the hypothesis of the above claim, and so we can induct. QED

Claim 7 *If G has a cycle C such that $G-C$ is disconnected, then one can induct.*

PROOF. If every vertex of C has a neighbor outside C, then there must be two consecutive vertices on C with neighbors in different components of $G-C$. Thus we can apply Claim 5. Otherwise, assume there is a vertex on C that has no neighbor outside C. Then choose such a vertex v such that one of its neighbors on C does have a neighbor outside C. The path $C-v$ satisfies the conditions of Claim 5 since v is isolated by the path's removal. QED

Thomassen determined the graphs that have no separating cycle. In particular, Theorem 3.1a of [10] can be summarized as:

Theorem 8 [10] *If G is a 2-connected graph with minimum degree at least 3 and with no cycle C such that $G - C$ is disconnected, then G is one of: the complete graph K_n, the wheel W_n, the balanced or almost balanced complete bipartite graph possibly with some specific edges added, the prism on six vertices, or one of the 23 graphs given in Figure 1 of that paper.*

It can then be checked that all the graphs given in Theorem 8 have either a 3-cycle or a 6-cycle, and so allow induction by Claim 4. This completes the proof of Theorem 2.

3 Graphs with Maximum Isolation Number

A graph in the family $\mathcal{G}$ as given by Lemańska et al. [8] is defined as follows. There is a connected *base* graph. Associated with each vertex of the base graph, which we call a *hook*, one introduces one pendant, which is either K_2 or C_5. For a K_2-pendant, the hook is joined to one or two vertices. For a C_5-pendant, the hook is joined to one, or any two, or three consecutive vertices of the 5-cycle (that is, any nonempty set that is not a vertex cover). See Figure 1.

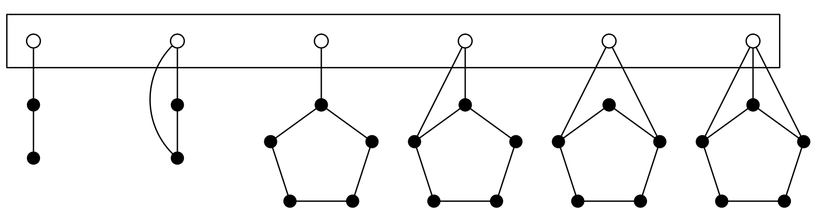

Figure 1: A graph in $\mathcal{G}$

Lemańska et al. [8] observed that these graphs have isolation number one-third their order. It is also immediate that something slightly stronger is true:

Lemma 9 *For a graph of $\mathcal{G}$ of order n, both its domination number and isolation number are $n/3$.*

We prove:

Theorem 10 *For $n \geq 15$, all connected graphs of order n with isolation number $n/3$ are in $\mathcal{G}$.*

For the proof it is helpful to determine all examples of equality. We define a family $\mathcal{E}$ of 14 exceptional graphs, shown in Figure 2. Note that the dashes represent optional edges, and thus some depictions represent multiple graphs. Up to isomorphism there are three graphs of order 6, eight graphs of order 9, and three graphs of order 12. (It is perhaps interesting to note that these graphs also have domination number equal to isolation number.)

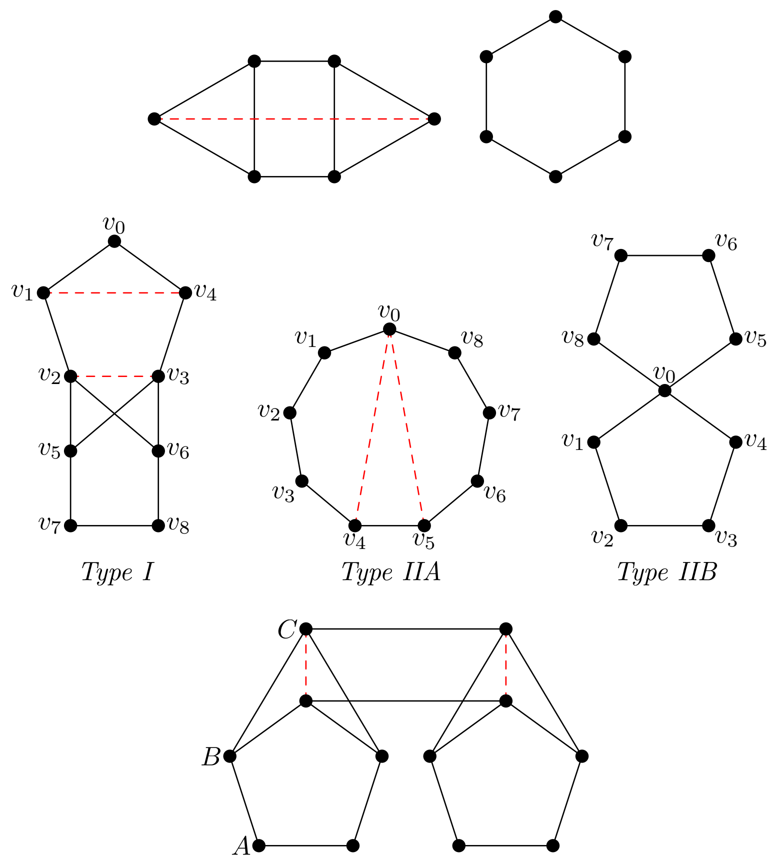

Figure 2: The graphs in $\mathcal{E}$

We prove the full characterization:

Theorem 11 *A connected graph has isolation number one-third its order if and only if it is in $\mathcal{G} \cup \mathcal{E}$.*

The proof is by induction on the order.

3.1 Preparing for Induction

The base case is orders 3, 6, and 9. The order-3 case is trivial and the order-6 case was noted in [8].

Lemma 12 *The connected graphs of order 9 with isolation number 3 are the 18 graphs in $\mathcal{G}$ and the 8 graphs in $\mathcal{E}$.*

Proof. By computer search. We enumerate all connected graphs of order 9 and determine which do not have an isolating set of cardinality 2. Java code is provided on https://people.computing.clemson.edu/~goddard/papers/isolation/. QED

For the inductive step, it is helpful to have a structural result. (This is somewhat similar to the idea used in the induction proof in [3].)

Lemma 13 *If G is a connected graph of order at least 3, then there exists a not necessarily induced star S on at least 3 vertices such that $G - S$ has at most one nontrivial component.*

Proof. Consider a spanning tree T of G. If T is a star, we are done; so assume T has diameter at least 3. Root T at the beginning of a longest path; say it starts at vertex a and ends with xyz.

If y has two or more children, then let S_1 be the star of y and all its children; since $T - S_1$ is connected it follows that $G - S_1$ is connected. So assume y has only one child. Note that we may similarly assume all children of x have degree either one or two in T.

If x has degree 2 in T, then consider the star S_2 of x, y, z; again $G - S_2$ is connected. So assume x has a second child. If there is an edge in G between grandchildren of x, say zz', then $G - S_3$ is connected where S_3 is the star of z, z', and y.

So assume there is no edge between grandchildren of x. Consider the star S_4 of x and all its children. Its removal from G leaves a component C containing vertex a, and potentially components containing grandchildren of x. But since there is no edge between grandchildren, each grandchild is either in a component by itself, or is joined by an edge to C. So S_4 is the desired star. QED

So assume G is a graph of order $n \geq 12$ with $\iota(G) = n/3$. By Lemma 13, there is a star S with at least three vertices such that $H = G - S$ has one nontrivial component and possibly some isolates. Say the star S is centered at x with neighbors $y_1, \ldots, y_d$. Then one can add x to the isolating set of H and so $\iota(G) \leq \iota(H) + 1$. Since H has $n - d - 1$ vertices, by Theorem 1 applied to the nontrivial component of H, it follows that $d = 2$, H has no isolate, and $\iota(H) = (n-3)/3$. By the induction hypothesis, the graph H is in $\mathcal{E} \cup \mathcal{G}$.

3.2 When H is in $\mathcal{E}$

Assume H is one of the exceptional graphs. We will need the following two "extendability" lemmas:

Claim 14 *If H is one of the exceptional graphs of order 12, then any two vertices of H can be extended to an isolating set of H of cardinality 4.*

Proof. Say vertices z_1 and z_2 are given. Call the degree-2 vertices that have a degree-2 neighbor *A-vertices*, the degree-3 vertices with an A-neighbor the *B-vertices*, and the remaining four vertices the *C-vertices*. We use "half" to mean one of the two components that would result if the edges joining C-vertices were to be removed. If z_1 and z_2 are from different halves, then let b_1 be a B-vertex distinct from z_1 and b_2 a B-vertex distinct from z_2; it can be checked that $\{z_1, z_2, b_1, b_2\}$ is an isolating set of H. If z_1 and z_2 are in the same half, there are two cases. If they are the two C-vertices, then add an A-vertex from each half. Otherwise, add a B-vertex and C-vertex from the other half, choosing a C-vertex whose neighbor in the first half is not already dominated if there is such a one. In both cases the quartet is an isolating set of H. QED

In the following, we use the labels and types as shown in Figure 2.

Claim 15 *If H is one of the exceptional graphs of order 9, then any two vertices of H can be extended to an isolating set of H of cardinality 3 except possibly pairs $\{v_0, v_1\}$, $\{v_0, v_4\}$, $\{v_7, v_8\}$ for type I, and pairs $\{v_2, v_3\}$, $\{v_6, v_7\}$ for type II.*

Proof. The following tables provide one possible third vertex for each pair, where in the second table, the notation a/b means vertex v_a works for type IIA and vertex v_b for type IIB.

Type I

	1	2	3	4	5	6	7	8
0	–	5	5	–	2	2	2	2
1		5	5	7	2	2	3	3
2			5	5	0	0	0	0
3				5	0	0	0	0
4					2	2	1	1
5						0	0	0
6							0	0
7								–

Type II

	1	2	3	4	5	6	7	8
0	5	5	5	5	1	2	3	4
1		6	6	6	0	2	3	4/0
2			–	7	0	0	4	5
3				8/7	0	0	0	5
4					0	0	0	0
5						1/2	2	2
6							–	2
7								3

QED

We continue with the overall proof. Recall that S has vertex set $\{x, y_1, y_2\}$ with x adjacent to y_1 and y_2.

Claim 16 *Vertex x has no neighbor in H.*

Proof. Suppose x has neighbor z in H. The vertex x can be added to any isolating set of $H - z$ to form an isolating set of G. Note that for all but one graph, an exceptional H is 2-connected. In that case, $H - z$ is connected, so its isolation number is at most $(n-4)/3$, and thus G is not extremal. A similar claim holds for the remaining possibility except when z is the cut-vertex thereof. But in that case $H - z$ has no component of order 2 or 5, and so again its isolation number is at most $(n - 4)/3$. QED

Claim 17 *Both y_1 and y_2 have neighbors in H.*

PROOF. Suppose that y_1 say has no neighbor in H. Then y_2 must have a neighbor in H, say z. Again $H-z$ has isolation number at most $(n-4)/3$, and an isolating set of $H-z$ can be extended to one of G by adding y_2, so that G is not extremal. QED

So assume y_1 has neighbor z_1 and y_2 has neighbor z_2 in H (possibly $z_1 = z_2$). Consider first the case that H has order 12. Then by Claim 14, one can extend $\{z_1, z_2\}$ to an isolating set J of H of size 4. The resultant J is also an isolating set of G, since only x remains of the star S; and thus G is not extremal.

Consider the case that H has order 9. Again, if $\{z_1, z_2\}$ can be extended to an isolating set J of H of cardinality 3, then the set J is an isolating set of G and hence G is not extremal.

Consider H of type II. Since by Claim 15 the two possible "bad" pairs are disjoint, the only possibility is that both y_1 and y_2 have unique neighbors in H. But if y_1 has neighbor v_2 and y_2 has neighbor v_3, then $\{v_1, v_6, y_2\}$ is an isolating set of G; and if y_1 has neighbor v_6 and y_2 has neighbor v_7, then $\{v_1, v_5, y_2\}$ is an isolating set of G. That is, the resultant G is not extremal. Consider H of type I. If both y_1 and y_2 have unique neighbors in H, then again it can be checked that the resultant G is not extremal. But if y_1 has neighbor v_0 and y_2 has neighbors v_1 and v_4, then for each H one obtains an extremal graph. However, note that the G with edge v_1v_4 present but not v_2v_3 is isomorphic to the G with edge v_2v_3 present but not v_1v_4. So one obtains that G is one of the three graphs of order 12 in $\mathcal{E}$.

3.3 When H is in $\mathcal{G}$

Assume H is in $\mathcal{G}$. Let B be the vertices of the base graph of H, and let C be a minimum isolating set of H that contains B. (It can easily be checked that one can choose such a minimum isolating set). For a pendant P, we use the notation $\widehat{P}$ for the set of its vertices and its hook.

Claim 18 *Vertex x has no neighbor in the pendants of H.*

PROOF. Suppose x has neighbor z in the pendants. Then vertex x can be added to any isolating set of $H-z$ to form one of G. Since $H-z$ has no component of order 2 or 5, its isolation number is at most $(n-4)/3$, and thus G is not extremal. QED

Claim 19 *We may assume vertices y_1 and y_2 are not adjacent.*

Proof. Suppose $y_1 y_2$ is an edge. Then, since one can choose x as any of the trio, it follows from Claim 18 that none of them has neighbors in the pendants. That is, their only neighbors outside S are in B. If two of them have neighbors in B, then C is an isolating set of G, a contradiction. Thus only one of the trio has a neighbor in B, and so G is in $\mathcal{G}$. QED

Claim 20 *We may assume both y_1 and y_2 have a neighbor in the pendants of H.*

Proof. Suppose y_2 say has no neighbor in the pendants. There are two cases:

Case 1: Assume y_2 has degree 1.

If all of y_1's neighbors are in B, then since C is not an isolating set of G, it follows that x has no neighbor in B; thus G is in $\mathcal{G}$. So assume y_1 has a neighbor in a pendant P with hook vertex v. Let $C' = C - \widehat{P}$ and consider deleting $N[C' \cup \{y_1\}]$ from G. Then only y_2 remains of S, and so all vertices outside P that remain are guaranteed to be isolated. Further, vertex v is adjacent to y_1 and so does not remain. If P was a K_2-pendant, then one isolated vertex of P remains; if P was a C_5-pendant, then what remains has a vertex a of degree 2 and removing $N[a]$ leaves no edges. That is, G has an isolating set of size $n/3 - 1$ and so is not an extremal graph, a contradiction.

Case 2: Assume y_2 has a neighbor in B.

If y_1 has a neighbor in B, then the set C would be an isolating set of G, a contradiction. So we may assume every neighbor of y_1 except x is in the pendants of H.

Case 2a) Assume y_1 has neighbors z_1 and z_2 in different pendants. Then deleting $\{y_1, z_1, z_2, x\}$ from G yields a graph G' with no 2- or 5-component except possibly the component containing B and y_2. But it is easily checked that if that component has order 5 then $|B| = 2$ and both pendants are K_2-pendants, which is impossible since $|H| \geq 9$. That is, $\iota(G) \leq (n-4)/3+1$, since y_1 can be added to any isolating set of G', and so G is not extremal.

Case 2b) Assume y_1 has neighbor(s) only in one pendant, say P with hook v. Let $C' = C - \widehat{P}$. Then removing $N[C'] - S$ from G yields isolates except for the component F induced by $V(P) \cup V(S)$. If P is a C_5-pendant, then F has order 8 and hence has an isolating set of two vertices. If P is a K_2-pendant, then F has order 5; but y_2 has degree 1 in F and so F is not a 5-cycle and hence has a 1-element isolating set. It follows that G is not extremal, a contradiction. QED

Claim 21 *If there exists one pendant P containing all neighbors of y_1 and y_2 in the pendants of H, then we are done.*

Proof. Let $C' = C - \widehat{P}$. Assume first that y_1 and y_2 have a common neighbor say z. Then C' can be extended to an isolating set C'' of H and hence G by adding z if P is K_2, or by adding z and a vertex of P otherwise; in either case $|C''| = |C| = (n-2)/3$, a contradiction.

Assume second that y_1 and y_2 have no common neighbor in P. Say z_1 is a neighbor of y_1 and z_2 is a neighbor of y_2. Then adding z_1 and z_2 to C' yields an isolating set of H and hence of G; so we are done unless P was a K_2. But in that case, one can easily check that since C is not an isolating set of G, adding the vertices of S to P turns it into a C_5-pendant. Thus G is in $\mathcal{G}$. QED

By Claims 20 and 21, we may assume that y_1 has a neighbor in some pendant P_1 with hook v_1, and y_2 has a neighbor in some pendant P_2 with hook v_2.

Claim 22 *If $|B| > 2$, then we are done.*

Proof. Assume first that there is another pendant of H containing a neighbor of y_1 or y_2. Say z_3 is a neighbor of y_1 in pendant P_3. Then consider the graph $G' = G - \{y_1, z_1, z_3, x\}$. We claim that G' has no component of order 2 or 5. To see this, observe that no vertex is removed from the base, and y_2 is in the component of G' containing the base. Removing z_1 or z_3 can separate a component in P_1

or P_3, but such a component has order 1 or 4. It follows by Theorem 1 that the isolation number of G' is at most $(n-4)/3$, and so G is not extremal.

So assume neither y_1 nor y_2 has a neighbor in any other pendant. Let $C' = C - (\widehat{P}_1 \cup \widehat{P}_2)$. Then, since $|B| > 2$, the set C' is not empty. If one removes from $H - (\widehat{P}_1 \cup \widehat{P}_2)$ the vertices of $N[C']$ therein, then all remaining vertices are from pendants, and they are now isolated. Further, none of the remaining vertices has a neighbor in $\widehat{P}_1$ or $\widehat{P}_2$, by the construction of graphs in $\mathcal{G}$; and none has a neighbor in S by the above discussion. That is, if one removes $N[C']$ from G, all remaining vertices outside $\widehat{P}_1 \cup \widehat{P}_2 \cup S$ are isolated. Further, since the base graph is connected, at least one of v_1 and v_2 is removed. Because S joins the two pendants, what remains of $\widehat{P}_1 \cup \widehat{P}_2 \cup S$ is connected, and its order is not a multiple of 3. It follows that $G - N[C']$ has isolation number less than one-third its order, which implies that G is not extremal. QED

Claim 23 *If $|B| = 2$, then we are done.*

Proof. Since H has order at least 9, there are two cases.

Case 1: Assume both P_1 and P_2 are C_5-pendants.
Then there is a vertex a_1 in P_1 such that $\{z_1, a_1\}$ dominates all of $\widehat{P}_1$ except possibly some vertex c_1 that has no neighbor in P_2. Similarly there is a vertex a_2 in P_2 such that $\{z_2, a_2\}$ dominates all of $\widehat{P}_2$ except possibly one vertex c_2 on the cycle. But the vertices c_1, c_2, x are independent. (Recall that by Claim 18 vertex x has no neighbor in the pendants.) Thus, the quartet $\{z_1, z_2, a_1, a_2\}$ is an isolating set of G of cardinality 4, a contradiction.

Case 2: Assume P_1 is a C_5-pendant and P_2 a K_2-pendant.
Then by the same reasoning as before, there is a vertex a_1 in P_1 so that $\{z_1, a_1\}$ dominates all of $\widehat{P}_1$ except possibly some vertex c_1, and c_1 has no neighbor in P_2. If one removes $N[\{z_1, a_1, z_2\}]$ from G, the only vertices that can remain are c_1, x, and v_2. Since the isolation number of G is 4, that removal must leave an edge. The only possibility for such an edge is xv_2; it follows thence that v_2 is not adjacent to z_2. See Figure 3.

Consider now the star $S' = P_2 \cup y_2$. Since y_1 has an edge to P_1, the graph $H' = G - S'$ is bridgeless and connected. So if H' is an extremal graph, it is in $\mathcal{E}$, and that case was already handled. QED

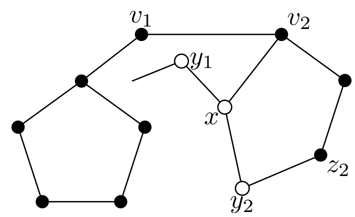

Figure 3: A portion of an extremal graph

This concludes the proof of Theorem 11.

4 Further Thoughts

We note that all graphs in $\mathcal{G}$ with nontrivial base graph have cut-vertices. So this raises the question of what is the asymptotic upper bound on the isolation number in 2-connected graphs. Another direction is to obtain better bounds for specific families of graphs, such as regular graphs (see [12]). Further, the relationship between domination and isolation deserves further investigation. For example, the corona of the complete graph has maximum domination number but minimum isolation number.

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